S&S QUIZ-9 SOLUTION

Sol(1):- Given signal,

$$x(t) = e^{-5t}u(t-t_0) + e^{-\beta t}u(t) \quad \text{---(1)}$$

Part-(1):- As we know that — $x(t) \underset{IFT}{\overset{FT}{\rightleftharpoons}} X(s)$

$$X(s) = \int_{-\infty}^{\infty} x(t)\cdot e^{-st}\,dt$$

$$= \int_{-\infty}^{\infty} e^{-5t}u(t-t_0)\,e^{-st}\,dt + \int_{-\infty}^{\infty} e^{-\beta t}u(t)\,e^{-st}\,dt$$

$$= \int_{t_0}^{\infty} e^{-(s+5)t}\,dt + \int_{0}^{\infty} e^{-(s+\beta)t}\,dt$$

$$= \frac{e^{-(s+5)t}}{-(s+5)}\Bigg|_{t_0}^{\infty} + \frac{e^{-(s+\beta)t}}{-(s+\beta)}\Bigg|_{0}^{\infty}$$

$$= \frac{e^{-\infty} - e^{-(s+5)t_0}}{-(s+5)} + \frac{e^{-\infty} - e^{-0}}{-(s+\beta)}$$

$$= \frac{e^{-(s+5)t_0}}{s+5} + \frac{1}{s+\beta} \rightarrow \text{(3 Marks)}$$

ROC: $\sigma > -5$ ROC: $\sigma > -\beta$ → (2X0.5 Marks)

For common ROC,

Case(I):- $\beta < 5$

Common ROC, $\sigma > -\beta$ → (1.5 Marks)

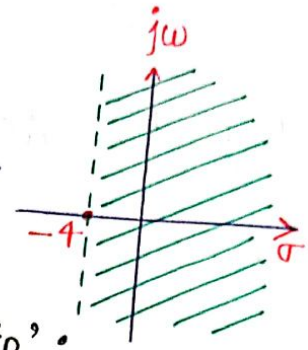
Scanned with OKEN Scanner

Case(II):- $\beta \geq 5$

Common ROC, $\sigma > -5$ → (1.5 Marks)

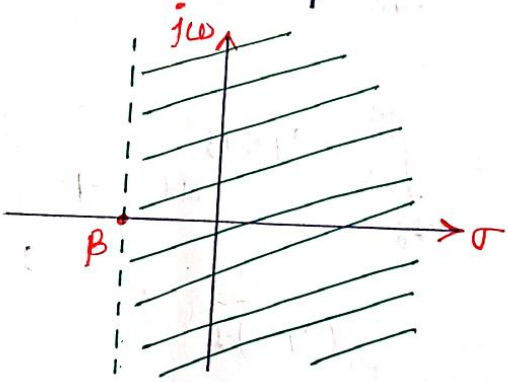
Part (2):- Given ROC of $X(s)$, $\text{Real}\{s\} > -4$

$\sigma > -4$



constraints on the real & imaginary part of 'β':

Case(I):- common ROC, $\sigma > -\beta$



Here, $\text{Real}\{\beta\} = 4$ → (2 Marks)

$\text{Img}\{\beta\}$ = No constraints → (1.0 Marks)

Scanned with OKEN Scanner